

DATA MINING AND MACHINE LEARNING APPLICATIONS

Edited By
Rohit Raja
Kapil Kumar Nagwanshi
Sandeep Kumar
K. Ramya Laxmi

Scrivener Publishing

WILEY

Data Mining and Machine Learning Applications

Scrivener Publishing
100 Cummings Center, Suite 541J
Beverly, MA 01915-6106

Publishers at Scrivener
Martin Scrivener (martin@scrivenerpublishing.com)
Phillip Carmical (pcarmical@scrivenerpublishing.com)

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This edition first published 2022 by John Wiley & Sons, Inc., 111 River Street, Hoboken, NJ 07030, USA and Scrivener Publishing LLC, 100 Cummings Center, Suite 541J, Beverly, MA 01915, USA
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Library of Congress Cataloging-in-Publication Data

ISBN 978-1-119-79178-2

Cover image: Pixabay.Com
Cover design by Russell Richardson

Set in size of 11pt and Minion Pro by Manila Typesetting Company, Makati, Philippines

Printed in the USA

10 9 8 7 6 5 4 3 2 1

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Preface

Data, the latest currency of today's world, is the new gold. In this new form of gold, the most beautiful jewels are data analytics and machine learning. Data mining and machine learning are considered interdisciplinary fields. Data mining is a subset of data analytics and machine learning involves the use of algorithms that automatically improve through experience based on data. However, the term data mining is a misnomer because it means to mine but not extract knowledge. A more apt term would be "knowledge discovery from data," since it is the practice of examining large pre-existing databases to generate information. Data mining algorithms are currently being investigated and applied worldwide.

Massive datasets can be classified and clustered to obtain accurate results. The most common technologies used include classification and clustering methods. Accuracy and error rates are calculated for regression and classification, and clustering to find actual results through algorithms like support vector machines and neural networks with forward and backward propagation. Applications include fraud detection, image processing, medical diagnosis, weather prediction, e-commerce and so forth. Data mining algorithms are even used to analyze data by using sentiment analysis. These applications have been increasing in different areas and fields. Web mining and text mining also paved their way to construct the concrete q2 field in data mining.

This book is intended for industrial and academic researchers, and scientists and engineers in the information technology, data science and machine and deep learning domains. Featured in the book are:

- A review of the state-of-the-art in data mining and machine learning,
- A review and description of the learning methods in human-computer interaction,

- Implementation strategies and future research directions used to meet the design and application requirements of several modern and real-time applications for a long time,
- The scope and implementation of a majority of data mining and machine learning strategies, and
- A discussion of real-time problems.

This book is a better choice than most other books available on the market because they were published a long time ago, and hence seldom elaborate on the current needs of data mining and machine learning. It is our hope that this book will promote mutual understanding among researchers in different disciplines, and facilitate future research development and collaborations.

We want to express our appreciation to all of the contributing authors who helped us tremendously with their contributions, time, critical thoughts, and suggestions to put together this peer-reviewed edited volume. The editors are also thankful to Scrivener Publishing and its team members for the opportunity to publish this volume. Lastly, we thank our family members for their love, support, encouragement, and patience during the entire period of this work.

Rohit Raja
Kapil Kumar Nagwanshi
Sandeep Kumar
K. Ramya Laxmi
November 2021